Nathaniel Morgan
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Curriculum Vitae

Education

Massachusetts Institute of Technology (MIT), Cambridge, MA — May. 2027
Candidate for Bachelor of Science | Major 6-3 Computer Science and Engineering — GPA: 4.5
Relevant Coursework: Deep Learning, Natural Language Processing, Cognitive Computation, Data Structures and Algorithms, Linear Algebra, Machine Learning, Probability & Statistics

Skills

Computer Science: Python, Java Script, Bash, SQLite,
DevOps: Google Cloud Platform, Kubernetes, Docker, Kubectl, Minikube, Tmux, Vim
Software: Google G-suite, Microsoft 365 Office, Adobe Creative Cloud
Languages: English, Spanish

Internships

Subconscious Dev Boston MA — May 2025 – Present
Founding AI Research Engineer
Assisted in the launch of Subconscious Dev through writing docs, testing platform, writing tests, modifying inference engine, and creating serverless deployment pipeline for RL-trained TIM + TIM RUN engine

- Containerized and modified fork of SGLang, implementing custom stop conditions and refining token handling and hidden state propagation for stable, high-quality long-horizon reasoning
- Created Python interface for Google DataCommons and AIME benchmarking of the modified inference engine and TIM model through constrained decoding pipelines
- Edited and contributed to the associated research paper (ICLR submission), ensuring accurate presentation of methods and results

BitEnergyAI, Cambridge MA — Jan 2025 - May 2025
Backend / Kernel Developer
Developed code for FP8 E4M3 multiplication on FPGA in C++ with Vitis-AI HLS to integrate transformer architecture with efficient floating point arithmetic operations.

- Designed host interface for multiplication between NxD weight matrix and DxM input matrix
- Implemented FP8 multiplication and addition through fused multiply add in C++
- Optimized matrix multiplication kernel through Vitis High-Level Synthesis best design practices
- Conducted software and hardware emulation of FPGA design

SALIERI.AI, Boston MA — May 2024 - Aug. 2024
Full-stack / NLP AI Engineer
Developed a server-less SaaS deployment with an autoscaling Kubernetes cluster hosted on GCP to facilitate structuring LLM output into parsable JSON based on business text document stores

- Utilized Node.js to create a web-hook API, enabling communication with a Python "child process" via inter-process communication (IPC).
- Developed 3 containerized micro-services for scalability and created network for intra-container communication
- Restricted LLM output using Context-Free Grammars in GBNF format and JSON to ensure valid generation (token generation constraint)
- Built and optimized an autoscaling Kubernetes cluster with MySQL data store on Google Cloud Platform, with containers stored in Google Artifact Registry and configured using Helm Chart for long-term sustainability.

Research

CSAIL, Cambridge MA — August 2025 – Present
Artificial Intelligence Researcher, Persistent Retrieval for Image-Semantic Memory **[**PRISM]
Researching persistent memory for Vision-Language Models (VLLMs) to overcome long-context limitations

- Evaluating methods on multimodal and long-horizon benchmarks (LoCoMo, MSC, KV, browser use)

CSAIL, Cambridge MA Jan 2025 – May 2025
Artificial Intelligence Researcher, Localized Optimization of Generative Information Contexts [LOGIC]
Leading research within the CSAIL Spoken Language Systems Group (SLS) at MIT, focusing on enhancing retriever performance in resource-augmented generation (RAG) systems through model editing.

- Benchmarked BERT-Large-Cased performance by generating a synthetic query-document set and extracting [CLS] token embeddings from a pretrained model using PyTorch
- Created vector database with FAISS Index and SQLite for efficient document storage and retrieval based on document embeddings
- Adapted MALMEN (ICLR 2025) to fit multi-class classification objective

CSAIL, Cambridge MA Sep 2024 – Dec 2024
Artificial Intelligence Researcher, Dynamic Data Storage and Retrieval [DDSR]
Lead research within the CSAIL Spoken Language Systems Group (SLS) at MIT into the utilization of Modern Continuous Hopfield Networks to model Memory for LLMs

- Constructed tensor-based data processing pipeline utilizing Kaggle sentence dataset and tokenization via. Mistral tokenizer
- Trained network with PyTorch and evaluated results with matplotlib

CSAIL, Cambridge MA Jan 2024 – May 2024
Artificial Intelligence Researcher, Thinking Deeper with Recursive Spawning [THREAD] Benchmark Creation
Conducted research within the CSAIL Spoken Language Systems Group (SLS) at MIT for the creation of a benchmark for LLM natural language embedded program generation.

- Produced >50 ground truth Python programs for synthetic data generation to create evaluation dataset.
- Created API calls for aggregate data collection and processing from Google Data Commons

Publications and Preprints

Phillip Schroeder, Nathaniel W. Morgan, Hongyin Luo, James R. Glass. May 2024
THREAD, "Thinking Deeper with Recursive Spawning" [NAACL 2025]
(arXiv preprint arXiv:2405.17402)

Hongyin Luo, Nathaniel W. Morgan, Tina Li, Derek Zhao, Ai Vy Ngo, et al. Jul 2025
Beyond Context Limits: Subconscious Threads for Long-Horizon Reasoning
(arXiv preprint arXiv:2507.16784)

Projects and Leadership

ARK (Automated Resource Knowledge Base), Open Source Project Lead Apr 2023 – Present
Leading a team of 6 across MIT, GT and UIC to redefine human computer interaction through a personalized agentic assistant interface

- Configured SGLang for LLM Inference and Hugging-Face TEI for Embeddings creation
- Coded agent runtime and connected it to state graph, long term memory, and MCP based tool-use module
- Organized project outline including "GitHub Org.", task list, weekly meetings, and various repositories for maintaining current and future contribution consistency as project scales
- Secured $5000 grant from MIT SIPB for development server build

Presentations

- SIPB Cluedumps: "AI for Dummies" Jan. 2024
- HackMIT Blueprint: "AI for Dummies" Mar. 2024
- TEDxBoston: "Visual Embedded Identification for Legitimacy" Jan. 2025

Awards & Accomplishments

MIT Student Information and Processing Board (SIPB) (Keyholder) Dec. 2023
Harvard x MIT CO-OP (Student Board Member) Aug. 2024
International Science and Engineering Fair (Finalist) Oct. 2021